

Amplitude-Guided Phase Masking for Hologram Compression

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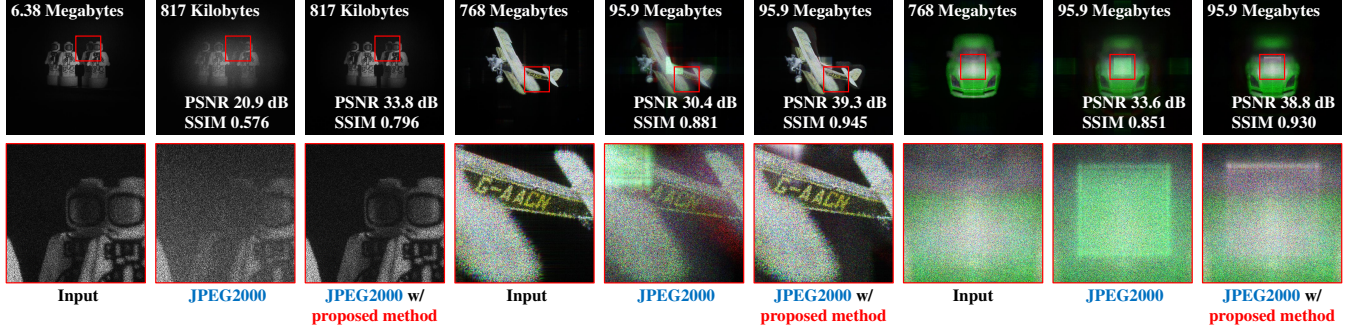


Figure 1: Comparison of 3D reconstructed images for Astronaut (left), Biplane (center), and SpecularCar (right), generated from phase information as input and compressed using JPEG2000 without and with our method.

Abstract

We introduce a hologram compression method that selectively preserves phase information in high-intensity regions using amplitude-guided masking. This approach improves coding efficiency while maintaining compatibility with existing image codecs, requiring no changes to encoders or decoders. Experiments demonstrate significant bitrate reductions and improved visual quality in shallow-depth scenes.

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1 Introduction

Holography records amplitude and phase of light, enabling full wavefront reconstruction for realistic 3D displays. However, phase information lacks spatial correlation, making it difficult to compress with existing image coding methods. We address this challenge by introducing amplitude-guided phase masking, which improves coding efficiency without sacrificing compatibility with existing image coding methods.

2 Proposed Method

Our method is developed based on the concept of object plane coding [Bernardo et al. 2018]. In holography, numerical propagation

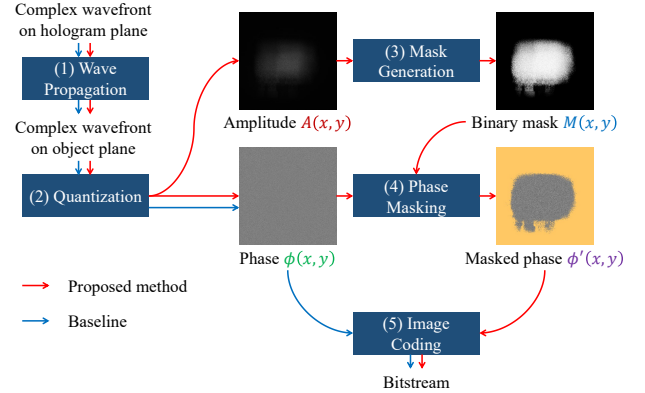


Figure 2: Workflow comparison of our method and baseline.

allows the reconstruction of wavefronts at arbitrary distances from the hologram. Object plane coding refers to the technique of encoding the wavefronts reconstructed at the object plane, which is the plane where the 3D object is in focus. The amplitude information on the object plane is known to exhibit signal characteristics similar to those of natural images, making it better suited for compression using existing image coding methods.

To improve the coding efficiency of phase information, we propose the amplitude-guided phase masking strategy, which preserves phase values in high optical intensity regions on the object plane. Our method focuses on compressing phase information. Compression of amplitude information is beyond the scope of this paper and is assumed to be handled separately. The process consists of five main steps as shown in Fig. 2. (1) The complex wavefront is numerically propagated to the object plane. (2) To enable compression using existing image coding methods, amplitude and phase information on the object plane are linearly quantized into integer-valued

images. (3) A binary mask is generated by comparing amplitude values against a threshold, distinguishing between regions that are expected to contribute significantly to the 3D reconstructed image and those that are not. (4) Phase values in regions excluded by the binary mask are replaced with a constant value, reducing entropy and improving compressibility. (5) The masked phase image is encoded using an existing image coding method.

This design of the mask generation and phase masking is based on our assumption that amplitude information can serve as a proxy for visual quality of the 3D reconstructed image. As illustrated in Fig. 2, our method preserves phase values in high amplitude regions, while masking those in low amplitude regions (visualized as yellow pixels) to increase redundancy and thereby reduce entropy. The detailed processing steps are described below.

Let $A(x, y)$ denote the amplitude at pixel (x, y) , and $\phi(x, y)$ the corresponding phase value. A binary mask $M(x, y)$ is defined as:

$$M(x, y) = \begin{cases} 1 & \text{if } A(x, y) \geq T_h, \\ 0 & \text{otherwise,} \end{cases}$$

where T_h is a predefined amplitude threshold. The masked phase image $\phi'(x, y)$ is then defined as:

$$\phi'(x, y) = \begin{cases} \phi(x, y) & \text{if } M(x, y) = 1, \\ E[\phi(x, y)] & \text{otherwise.} \end{cases}$$

Here, $E[\phi(x, y)]$ is the expected value of phase, approximated as the midpoint of the quantization range under the assumption of a uniform distribution.

On the decoder side, $\phi'(x, y)$ is decoded using an existing image decoder. The decoded image is combined with the amplitude image to reconstruct the complex wavefront. Numerical diffraction is then applied to reconstruct the hologram. Its computational complexity remains significantly lower than that of typical coding and display rendering pipelines [Bernardo et al. 2018].

Since our method masks phase information, some degradation in the quality of the 3D reconstructed image is inevitable. This trade-off is justified by the substantial improvement in coding efficiency, especially when data volume is a greater constraint than visual quality. Furthermore, our method works with existing image codecs without modifying encoders or decoders, enabling practical use in bandwidth- or storage-limited environments.

3 Experiments

To evaluate the effectiveness of our method, we conducted compression experiments using seven test holograms [Bernardo et al. 2018; Gilles et al. 2016]. We applied our method to two standardized image coding methods: HEVC and JPEG2000. Phase information was 8-bit quantized and encoded with and without our method. Threshold T_h was set to 16 experimentally.

Figure 1 presents 3D reconstructed images for scenes without background objects. In the Biplane and SpecularCar scenes, square-shaped bright artifacts appear in the center when using JPEG2000 alone. These artifacts occur because the wavefront, which should normally diffract through the aperture, is distorted and instead propagates without diffraction. In contrast, such artifacts are significantly reduced with our method. This is because more bits are allocated to phase information in regions with high optical intensity,

Table 1: BD-rate comparison for all test holograms. Negative values indicate better coding efficiency. The asterisks indicate cases where BD-rate could not be calculated. The blue asterisk (*) indicates test cases where the quality of our method at the highest compression ratio exceeded that of the baseline at the lowest compression ratio. The red double asterisk () indicates test cases where the RD-curves of our method and the baseline crossed.**

Hologram	BG	HEVC		JPEG2000	
		PSNR	SSIM	PSNR	SSIM
Astronaut	w/o	-78.1%	-76.9%	*	*
Biplane	w/o	-86.6%	-85.2%	*	*
SpecularCar	w/o	-81.8%	-87.9%	*	*
Horse	w/	-7.43%	-5.44%	-15.6%	-9.86%
DeepDices	w/	**	-6.46%	-14.6%	-14.2%
Ballet	w/	**	-3.35%	**	-5.79%
Breakdancers	w/	-22.2%	-20.2%	-41.3%	-26.0%

Note: Biplane, DeepDices, SpecularCar, Ballet, and Breakdancers provided by IRT b-com. Astronaut and Horse provided by [Bernardo et al. 2018].

allowing the wavefront to retain sufficient information for proper diffraction. Astronaut is an optically captured hologram, which tends to contain more speckle noise and is generally more difficult to compress than CGHs [Bernardo et al. 2018]. Nevertheless, our method improves coding efficiency even for Astronaut.

We objectively evaluated coding efficiency using Structural Similarity Index Measure (SSIM), Peak Signal-to-Noise Ratio (PSNR), and Bjøntegaard Delta (BD)-rate. BD-rate estimates average bitrate savings at equivalent quality and was calculated using QP levels of 39, 41, 43, and 45 for HEVC and compression ratios of 4, 5, 6, and 7 for JPEG2000.

BD-rate gain achieved by the proposed method varies depending on the presence of background objects in the scene. As shown in Table 1, our method achieved substantial bitrate reductions in scenes without background objects. For scenes with background objects, the results were more varied. Horse and Breakdancers showed moderate improvements, while DeepDices and Ballet exhibited inconsistent trends. In these cases, BD-rate could not be reliably calculated due to RD-curve crossings, particularly in HEVC for DeepDices and Ballet, and in JPEG2000 for Ballet.

4 Conclusions

We proposed an amplitude-guided phase masking for hologram compression. Our method improved coding efficiency in scenes without background objects, and also provided benefits in scenes with background objects, particularly those with shallow-depth. Future work includes developing adaptive thresholding strategies based on content characteristics.

Acknowledgments

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